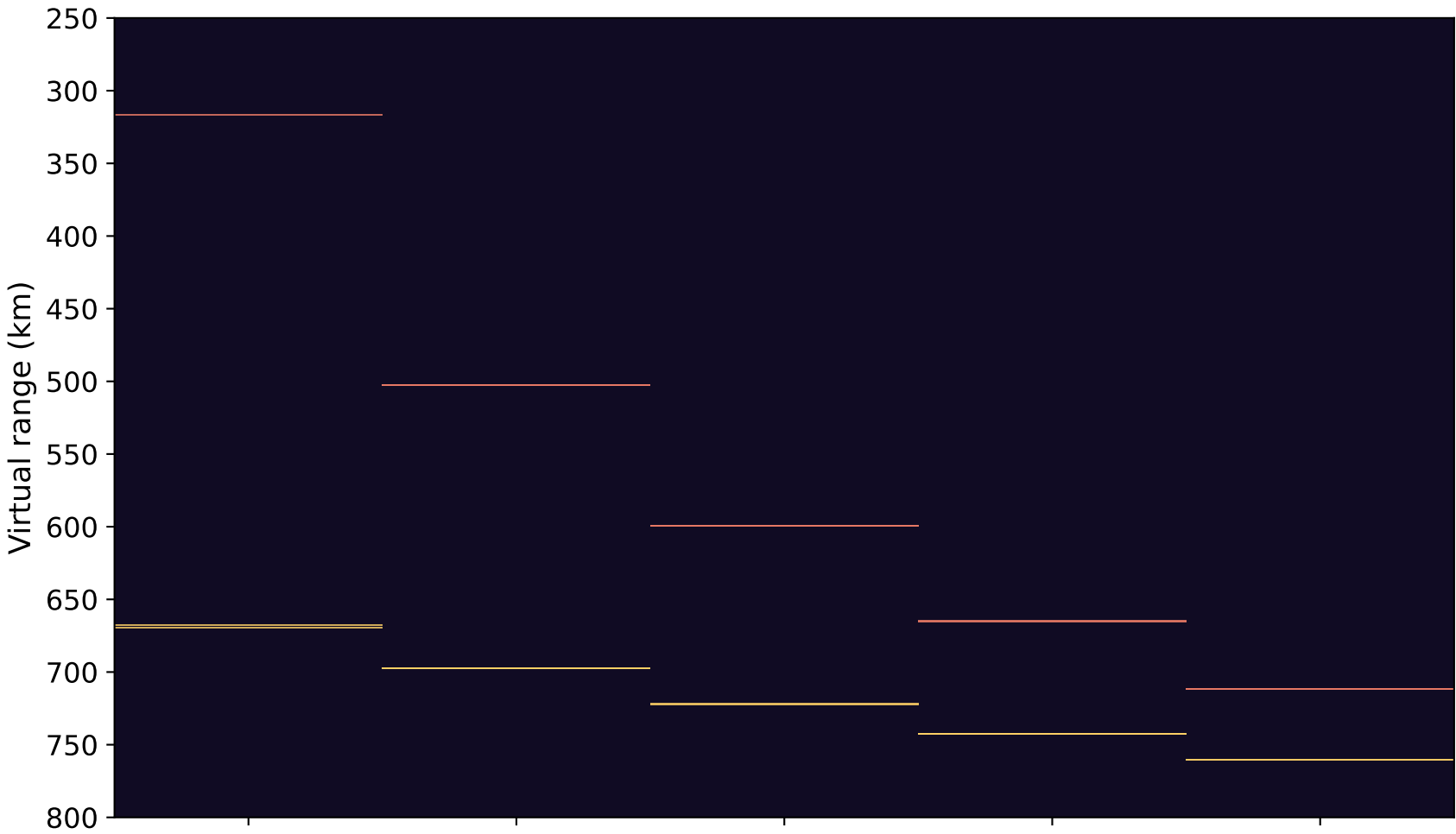


Shape-bump ionograms

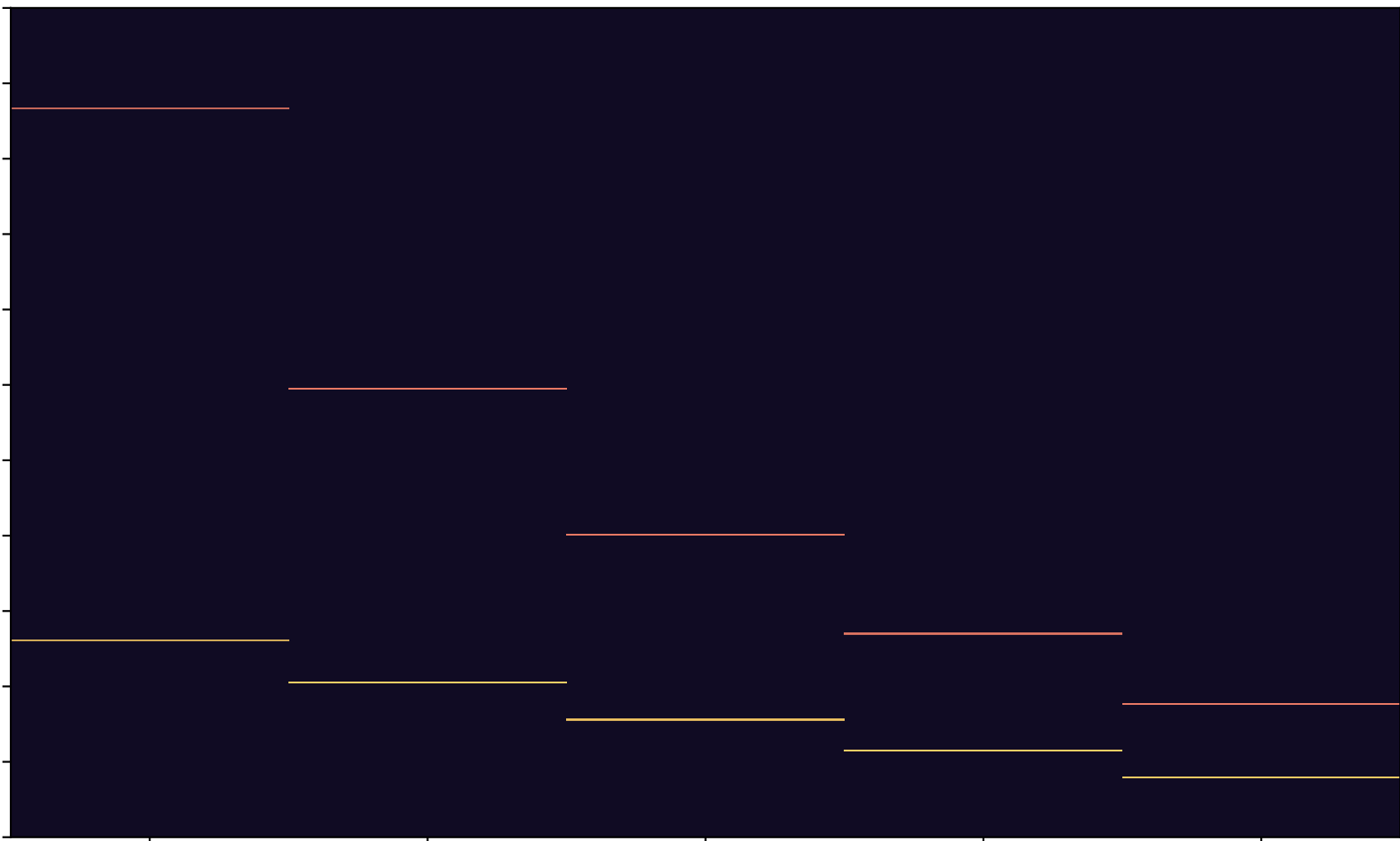
Synthetic ionosphere · 2.0–2.4 MHz window in 100 kHz steps · 1 km range bins · all accepted O/X returns

O mode X mode

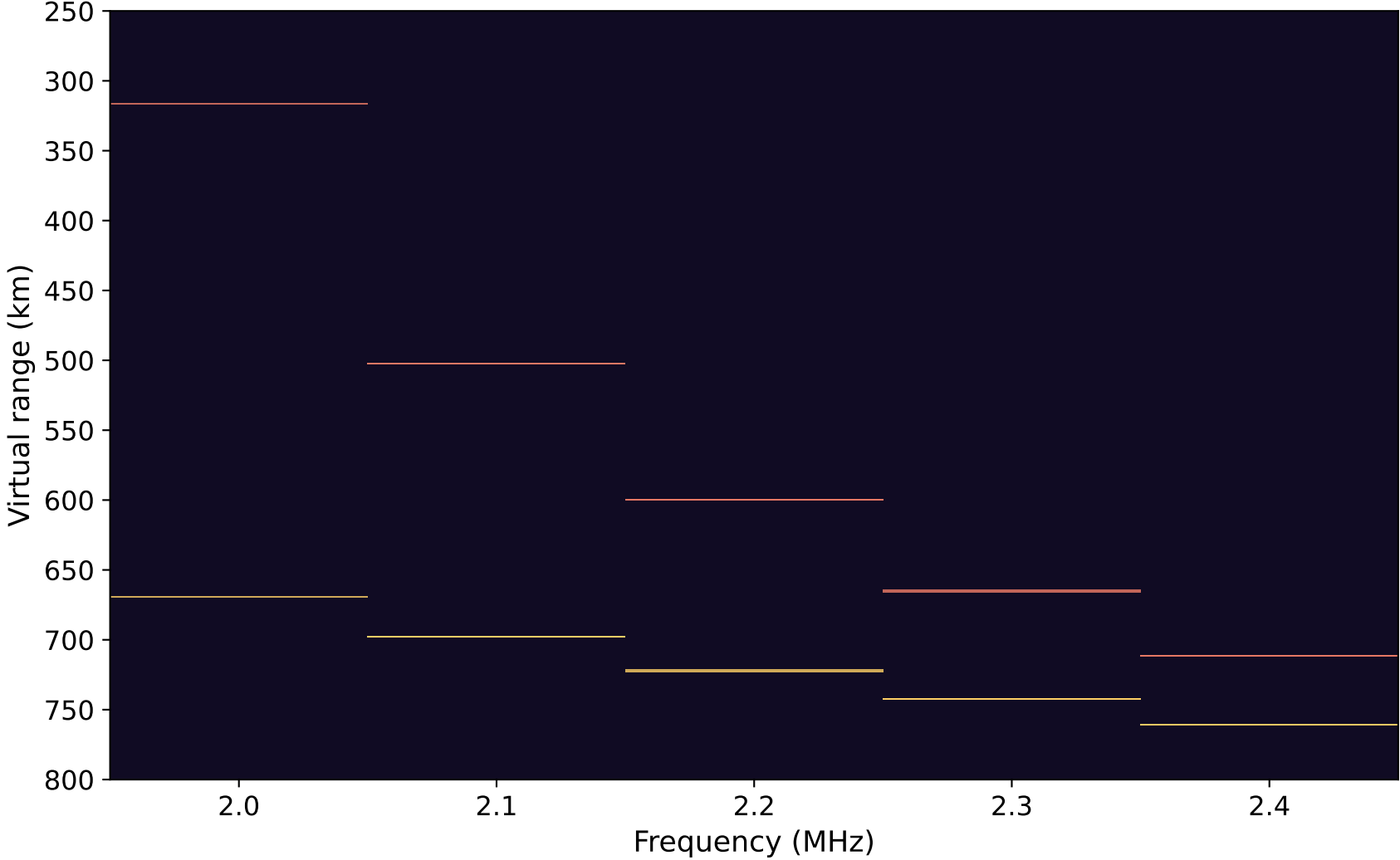
A Original az/el · 21 returns · 61.9 s
168 anchor / 288 full directions · reference



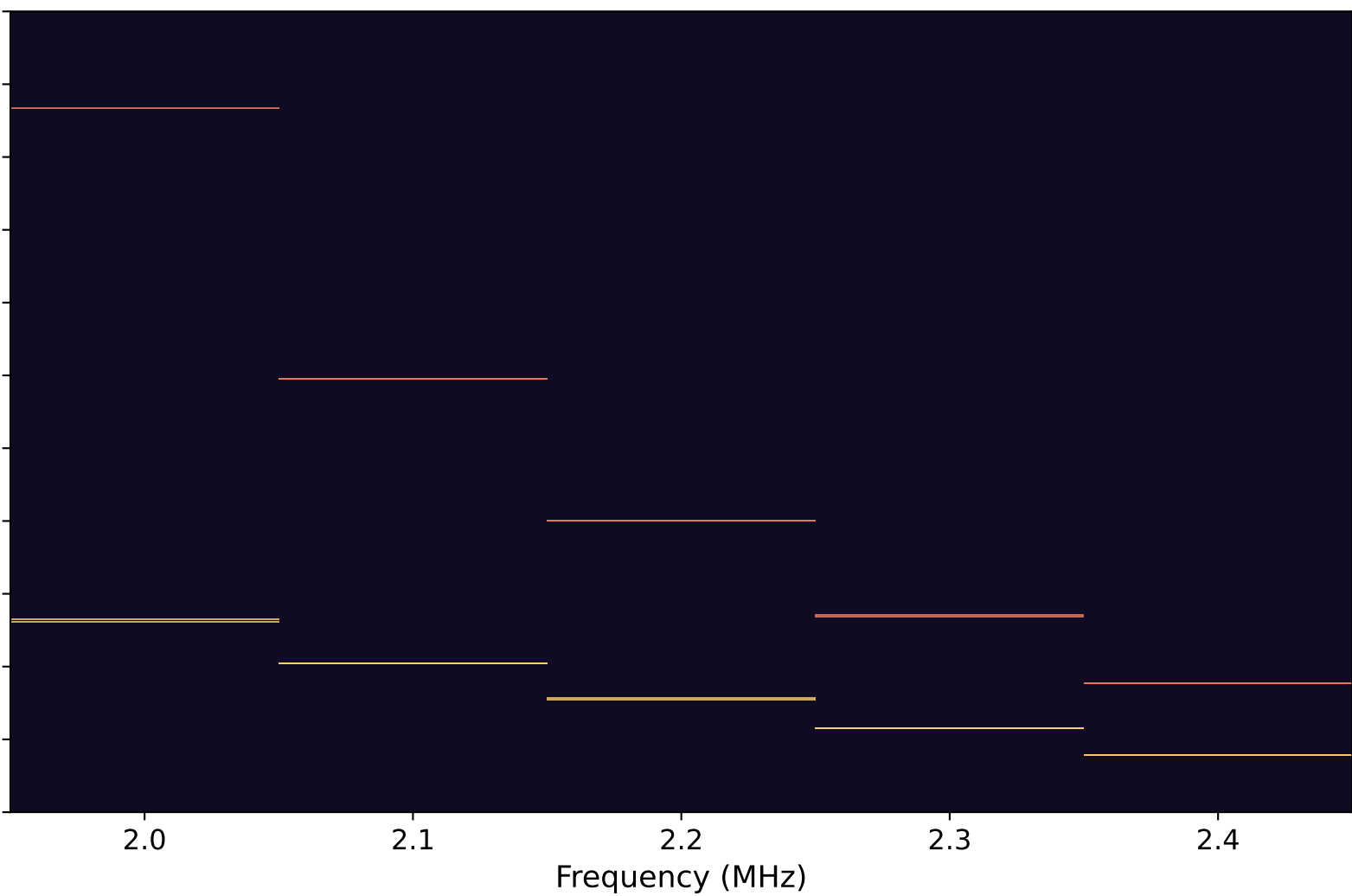
B Guarded full outer fan, 0 extra seeds · 20 returns · 44.5 s
174 anchor / 288 full directions · 20/21 reference returns matched



C Guarded half outer fan, 0 extra seeds · 20 returns · 41.7 s
105 anchor / 162 full directions · 20/21 reference returns matched



D Guarded half outer fan, 4 extra seeds · 21 returns · 57.4 s
105 anchor / 162 full directions · 21/21 reference returns matched



Homing gate: 1,000 m throughout. Match: same frequency and mode, virtual ranges within 1 km; coincident returns may blend. Timings cover both tracing modes; shared grid setup excluded.